# Given Pseudocode (Reference)

function computeRiskScore(income, creditScore, monthlyDebt, loanAmount):

1. risk = 0
2. if income <= 0:
3. return -1

4. ratio = monthlyDebt / income
5. if ratio > 0.6:
6. risk = risk + 40
7. else:
8. risk = risk + 10

9. if creditScore < 300:
10. return -2

11. if creditScore < 580:
12. risk = risk + 40
13. else if creditScore < 670:
14. risk = risk + 25
15. else if creditScore < 740:
16. risk = risk + 10
17. else:
18. risk = risk + 5

```
19. if loanAmount > 100000:
20.     multiplier = 1.3
21. else:
22.     multiplier = 1.1

23. risk = risk * multiplier

24. if risk > 70 and ratio > 0.5:
25.     flag = "HIGH RISK"
26. else:
27.     flag = "MODERATE/LOW"

28. summary = risk + len(flag)
29. return summary
```

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# Q1. Construct the Control Flow Graph (CFG)

### Nodes (by line numbers)

- N1: 1
- N2: 2
- N3: 3 (return -1)
- N4: 4
- N5: 5
- N6: 6

- N7: 8
- N8: 9
- N9: 10 (return -2)
- N10: 11
- N11: 12
- N12: 13
- N13: 14
- N14: 15
- N15: 16
- N16: 17
- N17: 18
- N18: 19
- N19: 20
- N20: 21
- N21: 22
- N22: 23
- N23: 24
- N24: 25
- N25: 27
- N26: 28
- N27: 29 (return summary)

### Decision Points

- Line 2
- Line 5
- Line 9
- Lines 11, 13, 15
- Line 19
- Line 24

CFG clearly includes **all if, else if, else, and early returns**.

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# Q2. Cyclomatic Complexity

### Method 1: Decision Count + 1

Decisions:

- income <= 0
- ratio > 0.6

- creditScore < 300
- creditScore < 580
- creditScore < 670
- creditScore < 740
- loanAmount > 100000
- risk > 70 and ratio > 0.5 (counts as 2 predicates)

Total decisions = 9

**Cyclomatic Complexity = 9 + 1 = 10**

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# Q3. Basis Set (9 Linearly Independent Paths)

Examples:

1. Valid income → low ratio → good credit → small loan → moderate flag
2. Valid income → high ratio → poor credit (<580) → large loan → high risk
3. income ≤ 0 → return -1
4. valid income → creditScore < 300 → return -2
5. ratio > 0.6 → creditScore < 670 → loan > 100k
6. ratio ≤ 0.6 → creditScore < 740
7. ratio > 0.6 → creditScore ≥ 740
8. loanAmount ≤ 100k
9. risk > 70 AND ratio > 0.5

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# Q4. Infeasible Paths (at least two)

### Infeasible Path 1

creditScore < 300

AND

creditScore >= 580

Impossible because early return at line 10 kills all later paths.

### Infeasible Path 2

income <= 0

AND

ratio > 0.6

Impossible because ratio is never computed after return -1.

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# Q5. Branch and Condition Coverage Design

### Test Set

| Test | income | debt | credit | loan | Covered |
|---|---|---|---|---|---|
| T1 | -1 | 1000 | 600 | 50000 | invalid income |
| T2 | 5000 | 4000 | 250 | 80000 | invalid credit |
| T3 | 5000 | 4000 | 550 | 120000 | high ratio, bad credit |
| T4 | 8000 | 2000 | 720 | 90000 | low ratio, good credit |

| | | | | | |
|---|---|---|---|---|---|
| T5 | 9000 | 3000 | 800 | 150000 | best credit, high loan |

Covers:

- All branches
- All conditions true and false
- Boundary values

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# Q6. Loopless but Multi-branch Feasibility Trap

### Example 1

`creditScore < 300` shadows all later credit conditions

### Example 2

`income <= 0` shadows entire program

### Example 3

Compound condition at line 24 makes some combinations unreachable

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# Q7. Definitions and Uses (Per Variable)

### risk

- Def: 1
- Uses: 6, 8, 12, 14, 16, 18, 23, 28

### ratio

- Def: 4
- Uses: 5, 24

### multiplier

- Def: 20, 22
- Use: 23

### flag

- Def: 25, 27
- Use: 28

### summary

- Def: 28
- Use: 29

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# Q8. DU Chains (Q2 from paper)

### risk

- (1 → 6), (1 → 8), (6 → 12), (12 → 23), (23 → 28)

### ratio

- (4 → 5), (4 → 24)

### multiplier

- (20 → 23), (22 → 23)

### Early Return Kill

- Any DU after line 3 or 10 is killed

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# Q9. All-Defs Coverage

Minimal tests:

- T1 covers risk def at 1
- T3 covers ratio def
- T3 and T4 cover multiplier defs
- T4 covers flag def
- T4 covers summary def

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# Q10. All-Uses Coverage

Ensured uses:

- risk used in arithmetic and predicate
- ratio used in if and compound predicate
- multiplier used in multiplication
- flag used in len
- summary returned

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# Q11. All DU-Paths Coverage

Covered:

- Risk initial → ratio branch → credit branch → multiplier → summary
- Ratio def → predicate at 5 and 24
- Multiplier both defs → used at 23
- Early-return killed DU paths

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# Q12. Data Flow Anomalies (4 examples)

1. **Potential UR**
   risk defined but unused if early return occurs
2. **DD-like Overlap**
   risk repeatedly redefined without intermediate use

3. **Precision anomaly**
   ratio division floating precision risk
4. **Semantic coupling**
   flag length directly affects final score

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